

Yaoyu Zhang

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324 Build. No.5, Science Buildings, Shanghai Jiao Tong University

Tenure-track Associate Professor, 2020. 9 –

Institute of Natural Sciences and School of Mathematical Sciences, Shanghai Jiao Tong University, Shanghai

Member, 2019. 9 – 2020. 7

School of Mathematics, Institute for Advanced Study, Princeton, New Jersey

Post-doctoral Associate, 2016-2019

NYUAD Institute, New York University Abu Dhabi and Courant Institute of Mathematical Sciences, New York University

Supervisors: David Cai, David W. McLaughlin

Education

Ph.D in Applied Mathematics, 2012-2016

School of Mathematical Sciences and Institute of Natural Sciences, Shanghai Jiao Tong University, Shanghai, China

Advisors: Douglas Zhou, David Cai

PhD thesis (in Chinese): *Sampling Artifacts of Granger Causality and the Reliable Reconstruction of Neuronal Network Connectivity*

B.S. in Applied Physics, minor in Applied Mathematics, 2008-2012

Zhiyuan College, Shanghai Jiao Tong University, Shanghai, China

Research Interests

Deep Learning Theory

- Condensation phenomenon
- Loss landscape structure
- Implicit regularization

Publications

A. Condensation phenomenon of deep learning

Condensation phenomenon: Neurons in the same layer tends to align with one another during the training

A1. Condensation phenomenon and its dynamical regime

1. Tao Luo, Zhi-Qin John Xu, Zheng Ma, Yaoyu Zhang, ["Phase Diagram for Two-layer ReLU Neural Networks at Infinite-Width Limit,"](#) Journal of Machine Learning Research (JMLR) 22(71):1–47, (2021).
2. Hanxu Zhou, Qixuan Zhou, Zhenyuan Jin, Tao Luo, Yaoyu Zhang, Zhi-Qin John Xu, ["Empirical Phase Diagram for Three-layer Neural Networks with Infinite Width,"](#) NeurIPS 2022.
3. Zhi-Qin John Xu, Yaoyu Zhang, Zhangchen Zhou, ["An overview of condensation phenomenon in deep learning,"](#) arXiv:2504.09484.

A2. Loss landscape structure—embedding principle series

1. Yaoyu Zhang, Zhongwang Zhang, Tao Luo, Zhi-Qin John Xu, ["Embedding Principle of Loss Landscape of Deep Neural Networks,"](#) NeurIPS 2021 spotlight.
2. Yaoyu Zhang, Yuqing Li, Zhongwang Zhang, Tao Luo, Zhi-Qin John Xu, ["Embedding Principle: a hierarchical structure of loss landscape of deep neural networks,"](#) Journal of Machine Learning, 1(1), pp. 60-113, 2022.
3. Hanxu Zhou, Qixuan Zhou, Tao Luo, Yaoyu Zhang, Zhi-Qin John Xu, ["Towards Understanding the Condensation of Neural Networks at Initial Training,"](#) NeurIPS 2022.
4. Zhiwei Bai, Tao Luo, Zhi-Qin John Xu, Yaoyu Zhang, ["Embedding Principle in Depth for the Loss Landscape Analysis of Deep Neural Networks,"](#) CSIAM Trans. Appl. Math., 5 (2024), pp. 350-389.
5. Leyang Zhang, Yaoyu Zhang, Tao Luo, ["Geometry of Critical Sets and Existence of Saddle Branches for Two-layer Neural Networks,"](#) arXiv:2405.17501 (2024).
6. Leyang Zhang, Yaoyu Zhang, Tao Luo, ["Uncovering Critical Sets of Deep Neural Networks via Sample-Independent Critical Lifting,"](#) arXiv:2505.13582 (2025).
7. Jiahan Zhang, Yaoyu Zhang, Tao Luo, ["Embedding principle of homogeneous neural network for classification problem,"](#) NeurIPS 2025.

A3. Generalization advantage—optimistic estimate series

1. Yaoyu Zhang, Zhongwang Zhang, Leyang Zhang, Zhiwei Bai, Tao Luo, Zhi-Qin John Xu, ["Linear Stability Hypothesis and Rank Stratification for Nonlinear Models,"](#) arXiv:2211.11623 (2022).
2. Yaoyu Zhang, Zhongwang Zhang, Leyang Zhang, Zhiwei Bai, Tao Luo, Zhi-Qin John Xu, ["Optimistic Estimate Uncovers the Potential of Nonlinear Models,"](#) arXiv:2307.08921 (2023).

3. Yaoyu Zhang, Leyang Zhang, Zhongwang Zhang, Zhiwei Bai, ["Local Linear Recovery Guarantee of Deep Neural Networks at Overparameterization,"](#) Journal of Machine Learning Research 26(69):1–30, 2025.
4. Tao Luo, Leyang Zhang, Yaoyu Zhang, ["Structure and Gradient Dynamics Near Global Minima of Two-layer Neural Networks,"](#) arXiv:2309.00508 (2023).

A4. Global dynamics and implicit bias

1. Leyang Zhang, Zhi-Qin John Xu, Tao Luo, Yaoyu Zhang, ["Limitation of Characterizing Implicit Regularization by Data-independent Functions,"](#) Transactions on Machine Learning Research (2023).
2. Zhiwei Bai, Jiajie Zhao, Yaoyu Zhang, ["Connectivity Shapes Implicit Regularization in Matrix Factorization Models for Matrix Completion",](#) NeurIPS 2024.
3. Jiajie Zhao, Zhiwei Bai, Yaoyu Zhang, ["Disentangle Sample Size and Initialization Effect on Perfect Generalization for Single-Neuron Target,"](#) arXiv:2405.13787 (2024).
4. Jiajie Zhao, Yaoyu Zhang, Tao Luo, ["Architecture Induces Structural Invariant Manifolds of Neural Network Training Dynamics",](#) arXiv:2510.09564v1 (2025).

A5. Condensation in language models

1. Zhongwang Zhang, Pengxiao Lin, Zhiwei Wang, Yaoyu Zhang, Zhi-Qin John Xu, ["Initialization is Critical to Whether Transformers Fit Composite Functions by Inference or Memorizing,"](#) NeurIPS 2024.
2. Zhiwei Wang, Yunji Wang, Zhongwang Zhang, Zhangchen Zhou, Hui Jin, Tianyang Hu, Jiacheng Sun, Zhenguo Li, Yaoyu Zhang, Zhi-Qin John Xu, ["The Buffer Mechanism for Multi-Step Information Reasoning in Language Models",](#) arXiv:2405.15302 (2024).
3. Zhongwang Zhang, Pengxiao Lin, Zhiwei Wang, Yaoyu Zhang, Zhi-Qin John Xu, ["Complexity Control Facilitates Reasoning-Based Compositional Generalization in Transformers",](#) arXiv:2501.08537 (2025).
4. Liangkai Hang, Junjie Yao, Zhiwei Bai, Tianyi Chen, Yang Chen, Rongjie Diao, Hezhou Li, Pengxiao Lin, Zhiwei Wang, Cheng Xu, Zhongwang Zhang, Zhangchen Zhou, Zhiyu Li, Zehao Lin, Kai Chen, Feiyu Xiong, Yaoyu Zhang, Weinan E, Hongkang Yang, Zhi-Qin John Xu, ["Scalable Complexity Control Facilitates Reasoning Ability of LLMs",](#) arXiv:2505.23013 (2025)

B. Frequency Principle of deep learning

Frequency Principle: neural networks tend to learn from low to high frequencies during the training.

1. **First Paper:** Zhiqin Xu, Yaoyu Zhang, Yanyang Xiao, ["Training Behavior of Deep Neural Network in Frequency Domain,"](#) ICONIP, pp. 264-274, 2019. (arXiv:1807.01251, Jul 2018)
2. **2021 World Artificial Intelligence Conference Youth Outstanding Paper Nomination Award:** Zhi-Qin John Xu, Yaoyu Zhang, Tao Luo, Yanyang Xiao, Zheng Ma, ["Frequency Principle: Fourier Analysis Sheds Light on Deep Neural Networks,"](#) CiCP 28(5). 1746-1767, 2020.
3. **Initialization effect:** Yaoyu Zhang, Zhi-Qin John Xu, Tao Luo, Zheng Ma, ["A Type of Generalization Error Induced by Initialization in Deep Neural Networks,"](#) MSML 2020.
4. **Linear Frequency Principle:** Yaoyu Zhang, Tao Luo, Zheng Ma, Zhi-Qin John Xu, ["Linear Frequency Principle Model to Understand the Absence of Overfitting in Neural Networks,"](#) Chinese Physics Letters (CPL) 38(3), 038701, 2021.
5. Tao Luo, Zheng Ma, Zhi-Qin John Xu, Yaoyu Zhang, ["Theory of the Frequency Principle for General Deep Neural Networks,"](#) CSIAM Trans. Appl. Math. 2 (2021), pp. 484-507.
6. **Linear Frequency Principle:** Tao Luo, Zheng Ma, Zhi-Qin John Xu, Yaoyu Zhang, ["On the exact computation of linear frequency principle dynamics and its generalization,"](#) SIAM Journal on Mathematics of Data Science 4 (4), 1272-1292, 2022.
7. **Minimal decay in frequency domain:** Tao Luo, Zheng Ma, Zhiwei Wang, Zhi-Qin John Xu, Yaoyu Zhang, ["An Upper Limit of Decaying Rate with Respect to Frequency in Deep Neural Network,"](#) MSML 2022.
8. **Overview:** Zhi-Qin John Xu, Yaoyu Zhang, Tao Luo, ["Overview Frequency Principle/Spectral Bias in Deep Learning,"](#) Communications on Applied Mathematics and Computation (2024): 1-38.
9. Zhangchen Zhou, Yaoyu Zhang, Zhi-Qin John Xu, ["A rationale from frequency perspective for grokking in training neural network,"](#) arXiv:2405.17479 (2024).

C. Deep Learning for Science

- Junjie Yao, Yuxiao Yi, Liangkai Hang, Weinan E, Weizong Wang, Yaoyu Zhang, Tianhan Zhang, Zhi-Qin John Xu, ["Solving multiscale dynamical systems by deep learning,"](#) Computer Physics Communications (2025):109802.
- Zhiwei Wang, Yaoyu Zhang, Pengxiao Lin, Enhao Zhao, E. Weinan, Tianhan Zhang, Zhi-Qin John Xu, ["Deep Mechanism Reduction \(DeePMR\) Method for Fuel Chemical Kinetics,"](#) Combustion and Flame 261 (2024): 113286.

- Tianhan Zhang, Yuxiao Yi, Yifan Xu, Zhi X. Chen, Yaoyu Zhang, Weinan E, Zhi-Qin John Xu, ["A Multi-scale Sampling Method for Accurate and Robust Deep Neural Network to Predict Combustion Chemical Kinetics,"](#) Combustion and Flame, 245, 112319, 2022.
- Lulu Zhang, Zhi-Qin John Xu, Yaoyu Zhang, ["Data-informed Deep Optimization,"](#) PLoS ONE 17 (6), e0270191, 2022.
- Jihong Wang, Zhi-Qin John Xu, Jiwei Zhang, Yaoyu Zhang, ["Implicit Bias with Ritz-Galerkin Method in Understanding Deep Learning for Solving PDEs,"](#) CSIAM Trans. Appl. Math. 3(2), pp. 299-317, 2022.
- Zhiwei Wang, Yaoyu Zhang, Yiguang Ju, Weinan E, Zhi-Qin John Xu, Tianhan Zhang, ["A Deep Learning-based Model Reduction \(DeePMR\) Method for Simplifying Chemical Kinetics,"](#) arXiv:2201.02025 (2022).
- Lulu Zhang, Tao Luo, Yaoyu Zhang, Weinan E, Zhi-Qin John Xu, Zheng Ma, ["MOD-Net: A Machine Learning Approach via Model-Operator-Data Network for Solving PDEs,"](#) Communications in Computational Physics 32(2) 299-335 2022.
- Tianhan Zhang, Yaoyu Zhang, Weinan E, Yiguang Ju, ["DLODE: A Deep Learning-based ODE Solver for Chemistry Kinetics,"](#) AIAA Scitech 2021 Forum, 1139.

D. Computational Neuroscience

- Kang You, Ziling Wei, Jing Yan, Boning Zhang, Qinghai Guo, Yaoyu Zhang, Zhezhi He, VISTREAM: Improving Computation Efficiency of Visual Perception Streaming via Law-of-Charge-Conservation Inspired Spiking Neural Network. CVPR 2025.
- Jing Yan, Yunxuan Feng, Wei Dai, Yaoyu Zhang, ["State-dependent Filtering of the Ring Model,"](#) arXiv:2408.01817 (2024).
- Yaoyu Zhang, Lai-Sang Young, ["DNN-Assisted Statistical Analysis of a Model of Local Cortical Circuits,"](#) Scientific Reports 10, 20139, 2020.
- Yaoyu Zhang, Yanyang Xiao, Douglas Zhou, David Cai, ["Spike-Triggered Regression for Synaptic Connectivity Reconstruction in Neuronal Networks,"](#) Frontiers in Computational Neuroscience 11, 101, 2017.
- Yaoyu Zhang, Yanyang Xiao, Douglas Zhou, David Cai, ["Granger Causality Analysis with Nonuniform Sampling and Its Application to Pulse-coupled Nonlinear Dynamics,"](#) Physical Review E 93, 042217, 2016.
- Douglas Zhou, Yaoyu Zhang, Yanyang Xiao, David Cai, ["Analysis of Sampling Artifacts on the Granger Causality Analysis for Topology Extraction of Neuronal Dynamics,"](#) Frontiers in Computational Neuroscience 8, 75, 2014.

- Douglas Zhou, Yaoyu Zhang, Yanyang Xiao, David Cai, ["Reliability of the Granger Causality Inference,"](#) New Journal of Physics 16 (4), 043016, 2014.
- Douglas Zhou, Yanyang Xiao, Yaoyu Zhang, Zhiqin Xu, David Cai, ["Granger Causality Network Reconstruction of Conductance-Based Integrate-and-Fire Neuronal Systems,"](#) PloS One 9 (2), e87636, 2014.
- Douglas Zhou, Yanyang Xiao, Yaoyu Zhang, Zhiqin Xu, David Cai, ["Causal and Structural Connectivity of Pulse-coupled Nonlinear Networks,"](#) Physical Review Letters 111 (5), 054102, 2013.

Select Presentations

Invited Talks

- **Condensation sheds light to the mathematical foundation of deep neural networks**, Southeast University, Oct. 31. Nanjing
- **The Condensation Phenomenon of Deep Neural Networks**, Flatiron Institute, Feb. 14, 2025, New York.
- **The Condensation Phenomenon of Deep Neural Networks**, MaD Seminar, New York University, Feb. 13, 2025, New York
- **The Condensation Phenomenon of Deep Neural Networks**, AMCS Colloquium, University of Pennsylvania, Feb. 10, 2025, Philadelphia
- **Optimistic Sample Size Estimate for Deep Neural Networks**, OneWorld Seminar, Oct. 9, 2024, Online
- **Optimistic Estimate of Sample Efficiency for Neural Networks**, Fudan University, Sep. 26, 2024, Shanghai
- **Optimistic Sample Size Estimate for Deep Neural Networks**, Numerical Analysis Seminar, The University of Hong Kong, Sep. 4, 2024, Online
- **Optimistic Estimate for Nonlinear Regression**, Tsinghua Statistics Seminar, Jun.28, 2023, Beijing
- **Embedding Principle of Loss Landscape of Deep Neural Networks**, 海纳百川上海青年学者讲座, 华为上海研究所, 2022.3.24, online
- **Embedding Principle of Loss Landscape of Deep Neural Networks**, AI TIME 青年科学家讲座, AI TIME, 2022.2.24, online
- **Embedding Principle of Loss Landscape of Deep Neural Networks**, Oct.16, 机器学习联合研讨计划, online
- **Impact of initialization on generalization of deep neural networks**, Oct.08, 2020, Math Machine Learning Seminar MPI MIS + UCLA, ZOOM.
- **A Fourier Analysis Perspective of Training Dynamics of Deep Neural Networks**, Dec.02, 2019, CMSA Special Seminar, Harvard University
- **A Fourier Analysis Perspective of Training Dynamics of Deep Neural Networks**, Nov.26, 2019, Seminar on Theoretical Machine Learning, Institute for Advanced Study, Princeton

- **Quantitative Understanding of F-Principle**, Jul.24, 2019, Beijing Computational Science Research Center
- **Validity and reliability of Granger Causality analysis on neuronal network reconstruction**, Nov.02, 2018, Purdue University
- **Characterization of the training process of neural network in phase space**, Oct.23, 2018, Rensselaer Polytechnic Institute
- **Validity and Reliability of Granger Causality Analysis on Neuronal Network Reconstruction**, Workshop on Data Assimilation and Information Theory, Jun.7, 2017, Fudan University
- **Spike-Triggered Regression on Neuronal Network Reconstruction**, Biomathematics/Computational Biology Colloquium Seminar, Nov.22, 2016, Courant Institute of Mathematical Sciences
- **Reliability of Granger Causality Analysis and Spike-triggered Regression method on Neuronal Network Reconstruction**, Dec.03, 2015, Rensselaer Polytechnic Institute
- **Validity and Reliability of Granger Causality Analysis on Neuronal Network Reconstruction**, Quantitative Biology Seminar, Nov.23, 2015, University of Michigan

Conference Talks

- **Condensation sheds light to the mathematical foundation of deep neural networks**, Workshop on Computational Harmonic Analysis and Data Science, Nov. 30-Dec. 5, 2025, Kunming
- **Condensation Sheds Light to the Mathematical Foundation of Deep Neural Networks**, The Mathematics of Scientific Machine Learning and Digital Twins, Nov. 22, 2025, Erice, Italy
- **The Condensation Phenomenon of Deep Learning**, SIAM Annual Meeting, Jul. 28, 2025, Montreal, Canada
- **Condensation sheds light to the mathematical foundation of deep neural networks**, Recent Trends in Applied Mathematics and Machine Learning, Jul.25-27, 2025, Jilin
- **Towards Understanding the Condensation Phenomenon of Deep Neural Networks**, IMS-NTU joint workshop on Applied Geometry for Data Sciences Part II, Jun. 2-6, 2025, Singapore
- **The Condensation Phenomenon of Deep Neural Networks**, MLPDE workshop, FAU MoD, Apr. 29, 2025, Erlangen, Germany
- **Optimistic Sample Size Estimate for Deep Neural Networks**, The 14th AIMS Conference, Dec.17, 2024, Abu Dhabi
- **Optimistic Sample Size Estimate for Deep Neural Networks**, Workshop on Mathematics of Machine Learning, NYU Shanghai, Nov. 24, 2024, Shanghai
- **Optimistic Sample Size Estimate for Deep Neural Networks**, NUS-PKU-SJTU Workshop, NUS, Nov. 9, 2024, Singapore
- **Optimistic Estimate of Sample Efficiency for Neural Networks**, 数学与人工智能科学大会 minisymposium, Aug. 29, 2024, Chongqing

- **Optimistic Estimate of Sample Efficiency for Neural Networks**, CSML2024, Aug. 10, 2024, Shanghai
- **Optimistic Estimate for Deep Neural Networks**, SIAM Annual Meeting minisymposium, Jul. 20, 2024, Online.
- **Understanding Condensation via Embedding Principle and Optimistic Estimate of Deep Neural Networks**, EASIAM minisymposium, Jun. 30, 2024, Macao, China
- **Embedding Principle: A hierarchical structure of deep learning loss landscape**, ICIAM minisymposium, Aug.24, 2023, Tokyo
- **Learning behavior of artificial neural networks**, 中国神经科学年会 (CNS2022), Nov.29,2022.online
- **Embedding Principle of Loss Landscape of Deep Neural Networks (Poster)**, DEEPMATH, Nov.4, 2021, online
- **Embedding Principle of Loss Landscape of Deep Neural Networks**, 人工智能的数学基础研讨会, Oct.16, Tong Ji University, Shanghai
- **Embedding Principle of Loss Landscape of Deep Neural Networks**, 机器学习与科学计算学术研讨会, Aug.9, 2021, Wuhan
- **Impact of initialization on generalization of deep neural networks**,第二届人工智能的数理基础青年论坛, Dec.05, Beijing
- **Phase diagram for two-layer ReLU neural networks at infinite-width limit**. Nov.01, CSIAM, Changsha
- **Dynamics of Deep Neural Networks – A Fourier Analysis Perspective**, Oct.4, 2019, Institute for Advanced Study, Princeton
- **Explicitizing an Implicit Bias of the Frequency Principle in Two-layer Neural Networks (Poster)**, DEEPMATH, Nov.11, 2019, New York City
- **A Novel Method on Neuronal Network Reconstruction**, 4th Statistical Physics and Complex System Conference, Jul.18, 2017, Xi'an, Shanxi
- **Spike-triggered Regression on Neuronal Network Reconstruction**, SIAM Conference on Dynamical Systems, May.25, 2017, Snowbird, Utah
- **Nonuniform Sampling Granger Causality Analysis and Its Application to Neuronal Network Reconstruction**, SIAM Conference on Nonlinear Waves and Coherent Structures, Aug.11, 2016, Philadelphia
- **Reliability of Granger Causality Analysis on Neuronal Network Reconstruction**, SIAM Conference on Life Sciences, Jul.11, 2016, Boston
- **Spike-triggered Regression method on Neuronal Network Reconstruction**, SIAM Conference on Life Sciences, Jul.11, 2016, Boston, Massachusetts

Event Organization

- **Theory and Application of Deep Learning Summer School**, Coorganizer, SJTU, Jul.3-7, 2023. Shanghai
- **Advances in Deep Learning Theory Minisymposia**, Organizer, SIAM Annual Meeting, Jul. 18-20, 2024. Online

- **The China Conference on Scientific Machine Learning 2024**, Chair of Organization Committee, Aug. 9-11, 2024. Shanghai
- **Advances in Deep Learning Theory Minisymposia**, Organizer, SIAM Annual Meeting, Jul. 28-Aug. 1, 2025. Montreal
- **The China Conference on Scientific Machine Learning 2025**, Member of Organization Committee, Aug. 8-10, 2025. Beijing
- **Advances in Deep Learning Theory Minisymposia**, Organizer, CSIAM Annual Meeting, Oct. 23-26, 2025 Changsha

Editorial

- Action Editor, *Neural Networks*.

Online Courses

Towards a Mathematical Foundation of Deep Learning: From Phenomena to Theory, FAU MoD Course, Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany, May 2-8, 2025.

- I. Mysteries of Deep Learning
- II. Frequency Principle/Spectral Bias
- III. Condensation Phenomenon
- IV. From Condensation to Loss Landscape Analysis
- V. From Condensation to Generalization Theory

Teaching Experiences

Courses

- Optimization methods (Graduate, MATH6015, Fall 2021)
- Statistical learning (Undergraduate MA451, Spring 2021)
- Mathematical analysis (Undergraduate MATH1214-01, MATH1608H-06, 2022-2026)
- Introduction to intelligence (Undergraduate AI1810-01, 2025)

Teaching assistant (with recitations):

- Biological physics (Undergraduate, Spring 2015)
- Methods of mathematical physics (Graduate, Fall 2013)
- Introduction to data science (Undergraduate, Summer 2013)
- Functional analysis (Undergraduate, Spring 2013)
- Complex analysis (Undergraduate, Fall 2012)

Honors

- Shanghai overseas high level talents program (2020)
- Qiu Suo outstanding young scholar program (2020)